ACTOR-CRITIC METHODS

INTERMEDIATE DEEP LEARNING

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PREVIOUSLY…

POLICY GRADIENT METHODS AND REINFORCE

PREVIOUSLY …

POLICY GRADIENT METHODS AND REINFORCE

- Policy gradient methods are a class of reinforcement learning algorithms that directly optimize the policy
- Estimate the gradient of expected return with respect to the policy parameters and updating them in the direction that increases expected reward.

$$\nabla_\theta J(\pi_\theta) = E_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T} R_t(\tau) \, \nabla_\theta \log \pi_\theta \left(a_t | s_t \right) \right]$$

PREVIOUSLY …

POLICY GRADIENT METHODS AND REINFORCE

- REINFORCE is a Monte Carlo-based policy gradient algorithm
- But it has some drawbacks :
  - High variance
  - Credit assignment problem
  - Inefficient updates (only after an episode finishes)



PREVIOUSLY …

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PREVIOUSLY …

POLICY GRADIENT METHODS AND REINFORCE

- REINFORCE is a Monte Carlo-based policy gradient algorithm
- But it has some drawbacks :
  - High variance
  - Credit assignment problem
  - Inefficient updates (only after an episode finishes)
- What if we can better evaluate the influence of action?



INTRODUCTION

ACTOR-CRITIC METHODS

ACTOR-CRITIC METHODS



Actor



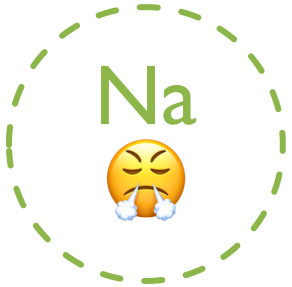
Critic

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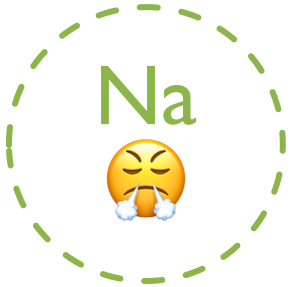






Actor







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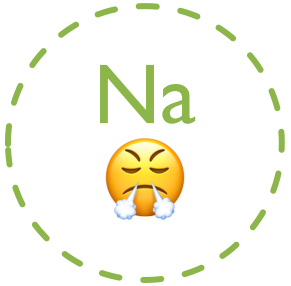






Actor







Critic

INTRODUCTION

ACTOR-CRITIC METHODS

Policy function







Actor

Actions

Yess

Na

Values

Value function



Critic

INTRODUCTION

ACTOR-CRITIC METHODS

- A learned reinforcement signal can be more informative for a policy than the raw rewards provided by the environment
- A common option is the advantage function:

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- Where $A_\pi(s,a)$: the advantage,
- $Q_\pi(s,a)$: the expected return of taking action a in state s
- $V_\pi(s)$: the value (~expected return) of state s when following the current policy

ADVANTAGE ACTOR-CRITIC (A2C)

ADVANTAGE – N-STEPS RETURNS – GENERALIZED ADVANTAGE ESTIMATION (GAE)

ADVANTAGE FUNCTION

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- Intuition: $A_\pi(s,a)$ measures how much better or worse an action is compared to the average action chosen by the policy in that state.
- If an action is worse than the average but still receives a positive return?
  - E.g.

$$Q_\pi(s,a) = 90, \quad V_\pi(s) = 100, \quad A_\pi(s,a) = -10$$

or

$$Q_\pi(s,a) = -90, \quad V_\pi(s) = -100, \quad A_\pi(s,a) = 10$$

- If the policy is already good?

ADVANTAGE ACTOR-CRITIC (A2C)

ADVANTAGE FUNCTION

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- It's a "baseline-adjusted" estimate of an action's influence → helps reduce variance in learning.
- It can capture long-term effects of the current action without directly attributing the outcome to previous actions → focuses on what matters now
- But how to approximate this function?

ADVANTAGE FUNCTION

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- It's the critic's job to estimate the values used in the advantage function.
- One critic for $Q_\pi(s,a)$ and one for $V_\pi(s)$

$Q_\pi(s,a)$





$V_\pi(s)$

ADVANTAGE FUNCTION

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- It's the critic's job to estimate the values used in the advantage function.
- One critic for $Q_\pi(s,a)$ and one for $V_\pi(s)$?
  - Less efficient to train
  - Two estimates should be consistent

$Q_\pi(s,a)$  $V_\pi(s)$

ADVANTAGE ACTOR-CRITIC (A2C)

N-STEP RETURNS

- Typically, we learn just V_π and combine it with reward to estimate Q_π.
  - Q_π is more complex and needs more samples
  - Using Q_π to estimate V_π is more computationally expensive

ADVANTAGE ACTOR-CRITIC (A2C)

N-STEP RETURNS

- The goal is to use $Q_\pi(s,a)$ and $V_\pi(s)$ to approximate return G
- Reminder: $G_t = \sum_{k=t}^{T} \gamma^{k-t} r_k$
- If

$$V_\pi(s_t) = E_\pi[\, G_t \mid s_t \,], \qquad Q_\pi(s_t, a_t) = E_\pi[\, G_t \mid s_t, a_t \,]$$

- Then:

$$Q_\pi(s_t, a_t) = E[\, r_t \mid s_t, a_t \,] + \gamma E_\pi[\, G_{t+1} \mid s_{t+1} \,]$$

$$Q_\pi(s_t, a_t) \approx r_t + \gamma V_\pi(s_{t+1})$$

- Which is a one−step approximation (sounds familiar?)

ADVANTAGE ACTOR-CRITIC (A2C)

N-STEP RETURNS

- We can expand to more steps: each sampled return provides an unbiased sample of the expected return.
- But going too far into the future increases variance
- So we use an n-step return:

$$Q_\pi(s_t, a_t) \approx r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots + \gamma^{n-1} r_{t+n-1} + \gamma^n V_\pi(s_{t+n})$$

- n controls the trade-off between bias and variance
  - When $n = 1$?
  - When $n = T - t$?

ADVANTAGE ACTOR-CRITIC (A2C)

N-STEP RETURNS

- Substituting back into

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- We obtain:

$$A_\pi^{\text{n-step}}(s_t,a_t) \approx r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots + \gamma^{n-1} r_{t+n-1} + \gamma^n V_\pi(s_{t+n}) - V_\pi(s_t)$$

- Which is the formula of advantage estimated with n-steps return.

ADVANTAGE ACTOR-CRITIC (A2C)

GENERALIZED ADVANTAGE ESTIMATION (GAE)

- With n-step returns, we are already balancing bias and variance by fixing n.
- However,
  - Choosing n is difficult
  - The trade-off is *abrupt or discontinuous* (it changes at a specific step)
- Is there a way to make this balance softer and smoother?

GENERALIZED ADVANTAGE ESTIMATION (GAE)

- Generalized Advantage Estimation (GAE) was proposed as an advantage estimator that provides a continuous trade-off between bias and variance.

$$A_{\pi}^{\mathrm{GAE}}(s_t, a_t) = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}$$

- Where

$$\delta_t = r_t + \gamma V_{\pi}(s_{t+1}) - V_{\pi}(s_t), \qquad \lambda \in [0,1]$$

- Intuitively, each TD error measures how "surprising" a step is, and λ controls how far into the future these surprises still matter

GENERALIZED ADVANTAGE ESTIMATION (GAE)

$$A_{\pi}^{\mathrm{GAE}}(s_t, a_t) = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}$$

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$$\delta_t = r_t + \gamma V_{\pi}(s_{t+1}) - V_{\pi}(s_t), \qquad \lambda \in [0,1]$$

- Where does this λ come from?
- The idea behind GAE is to combine all n-step returns for $n \in \{1,2, \dots\}$…

ADVANTAGE ACTOR-CRITIC (A2C)

GENERALIZED ADVANTAGE ESTIMATION (GAE)

- Idea: GAE can be viewed as an exponentially weighted average (EWA) of all the n-step advantage estimates.
- Given a sequence $(X_1, X_2, \dots, X_t, \dots)$, the exponentially weighted average of X is:

$$Y_t = \lambda Y_{t-1} + (1 - \lambda) X_t$$

